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Volume 2 Quick Reference Guide

Solar Home Systems: PAYGo formulas, definitions, defaults and checklists

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How to use this guide

This guide condenses Volume 2 into reference cards for the desk: the formulas, definitions, defaults, thresholds and checklists that a PAYGo analyst, lender or investor needs within reach. Each card stands alone. The reasoning behind every rule is in the book; the calculations are in the model and the tools.

Product	Use it for	File
Book	The reasoning: why each number matters and how to read it	AEF_Volume2_SHS_PAYGo_Book_Draft_v0.1.pdf
Model	The integrated five year plan, credit engine, covenants, valuation, readiness gates	AEF_SHS_PAYGo_Model_v0.7.xlsx
User manual	Twelve step workflow through the model; methodology reference	AEF_SHS_PAYGo_User_Manual_v0.7.pdf
Case study	SolaraPay Ltd: an end to end investment decision on synthetic data	SolaraPay_Case_Study_v0.7.pdf and case workbook
Templates	Memo, diligence checklist, credit policy, agent economics, lender report, loan tape, borrowing base, canvas	templates/AEF_V2_T01 to T08
Decision tools	Price plan and APR, unit economics, calibration, financing, screening, returns	decision-tools/AEF_V2_D1 to D6

All example figures refer either to the book's illustrations or to the fictional SolaraPay case, and are flagged as such. Default model inputs describe a fictional market. Negative figures are shown in parentheses.

The PAYGo business on one page

A PAYGo solar company is three businesses in one legal entity: a retailer that books hardware margin at sale, a service provider that keeps the device working for two to four years, and a consumer lender that waits for the cash. Its income statement can show a profit while it runs out of cash, so it must be read through cohorts and the receivables book, not through the sales line.

Read first	Why	Where
Cohort repayment against plan at the same age	Earliest honest signal of credit quality	Vintage_Dashboard; T05 Vintage
Collection rate, trailing 3 months, deposits excluded	The usual covenant measure	Credit_Portfolio; T05
PAR30, reported and lagged 3 months	Growth flatters the reported figure	T05 KPIs
Peak equity requirement in USD	The real funding need of the plan	Investment_Summary; D4
Investor IRR and multiple in USD, Base and Downside	The return after currency	Valuation; D6

How PAYGo companies fail

Failure mode	What it looks like first	Test
Growth outruns funding	Funding need rises faster than profit; facility delayed or in breach	Peak equity; months of runway; facility start date
Credit drift	Each new cohort a little below the last; portfolio ratios stable for a year	Cohort curves at M3, M6, M12 against plan
Currency	Hardware and debt in USD, collections in local currency	FX pass through; dollar debt share; depreciation sensitivity

None of the three shows first on the income statement. All three show first in the cohort curves, the funding plan and the FX bridge.

Core formulas

Quantity	Formula	Notes
Monthly instalment	Daily rate $\times 365 \div 12$	
Total contract value	Deposit + instalment $\times$ tenor	
PAYGo premium	Total contract value less cash price	Financing income, straight line over the tenor in the model
Amount financed	Cash price less deposit	
Implied monthly rate r	Solves amount financed = instalment $\times [1 - (1 + r)^{-T}] \div r$	Spreadsheet: RATE(T, -instalment, amount financed)
Nominal APR	$12 \times r$	State the convention used in any disclosure
Effective annual rate	$(1 + r)^{12} - 1$	
Flat rate	Premium $\div$ amount financed $\div (T \div 12)$	Understates the true cost
Payment burden	Instalment $\div$ monthly household income	Test on lean month income
Survival to age a	$S(a) = (1 - h)^a$, with h the monthly default hazard	Constant hazard
Collection at age a	Instalment $\times c \times S(a)$, with c the collection rate on paying accounts	
Expected share of instalments collected over the tenor	$c \times q \times (1 - q^T) \div (h \times T)$, with $q = 1 - h$	
Cumulative repayment at age m	$c \times q \times (1 - q^m) \div (h \times m)$, for m within the tenor	Vintage measure
Receivable at risk (proxy)	$(1 - S(a)) \times (T - a) \times (\text{instalment} - \text{premium} \div T)$	Carrying amount of accounts that stopped paying
Net recovery	Defaults at age $(a - \text{lag}) \times \text{repossession rate} \times \text{resale share} \times \text{cash price} \times (1 - \text{recovery cost})$	
ECL at origination (model)	Lifetime expected missed instalments	Consumed as instalments are missed
Borrowing base	Σ advance rate $\times$ eligible receivables, less concentration excess	Eligible: lower DPD bound at or below the maximum
Facility drawn (model)	$\max(0, \min(\text{limit}, \text{borrowing base}, \text{previous balance} + \text{minimum cash} - \text{cash before facility}))$	Cash sweep; no circularity
FX loss on USD debt	Opening USD balance $\times$ change in exchange rate	Through the income statement
Break even depreciation	$(1 + \text{local currency rate}) \div (1 + \text{dollar rate}) - 1$	Above it, dollar debt costs more
Terminal value (normalised)	$[(\text{EBITDA} - \text{tax} - \text{capex}) \times (1 + g) - g \times \text{NWC}] \div (r - g)$, Year 5 values	Avoids capitalising a growth year
Investor stake	Ticket $\div (\text{pre money valuation} + \text{ticket})$	
Investor IRR, single exit, no follow on	$(\text{stake} \times \text{exit equity in USD} \div \text{ticket})^{(1 \div n)} - 1$	Exit equity floored at zero
Maximum pre money for a target IRR	$\text{Exit equity in USD} \div (1 + \text{target})^n - \text{ticket}$	No follow on equity

Portfolio KPIs

KPI	Definition	Common error
Collection rate	Instalments collected ÷ instalments due, both excluding deposits	Deposits in the numerator
Trailing 3 month collection rate	Collections ÷ instalments due over 3 months	Covenant tested on a single month
PAR30, PAR90	Balance of accounts more than 30 (90) DPD ÷ gross receivables	Arrears instead of whole balances
PAR30 lagged 3 months	Balance more than 30 DPD ÷ gross receivables 3 months earlier	Not reported in fast growing books
Receivables at risk	Company definition, aligned with current PAYGo PERFORM documents	Definition not printed beside the figure
Write off ratio	Write offs (12 months) ÷ average gross receivables	Read without the write off policy
Recovery rate	Recoveries net of costs ÷ write offs	Assumed rather than observed
Repayment rate (cohort)	Cumulative collections ÷ cumulative instalments due at a given age	Comparing cohorts at different ages
Ownership rate	Share of a cohort owning its device at twice the tenor	Claimed without unlock evidence
Active ratio	Accounts paying in the last 30 days ÷ accounts not paid off or written off	
Enabled rate	Share of active accounts with credit at period end	Called "unlock rate" in operations; not the permanent unlock
DPD	Schedule shortfall in days of the daily rate	Days without credit used without disclosure

Every DPD distribution must add up to gross receivables. A report that does not reconcile is not used.

Numbers worth remembering

These are results from the book's illustrations and the default model, not sector standards. They show the size of the effects.

Effect	Illustration	Source
Survival on a 2.6% monthly hazard	72.9% still paying at month 12; 53.1% at month 24	Chapter 6
Tenor and collection	69.2% of instalments collected over 18 months against 64.4% over 24 months (hazard 2.6%, collection 88%)	Chapter 4
Growth and the collection rate	Same customers: 64.4% at zero growth, 68.3% at 5% a month, 71.6% at 10% a month	Chapter 6; D3
Growth and PAR30	New lending of LCY 400m on an LCY 800m book with 20% PAR30 shows 14.3% reported, 21.5% lagged	Chapter 7
Margin against cash	A Tier 2 sale booking LCY 14,000 of contribution at sale is LCY (22,000) in cash and pays back in month 14	Chapter 1; D2
Loss rate and value	Ten points more expected loss cut unit contribution by 41% and unit IRR from about 49% to about 28%	Chapter 9
Fully loaded CAC	About 1.86 times the narrow CAC in the Chapter 5 territory example	Chapter 5; T04
Borrowing base headroom	A 3.4% migration from 1 to 30 to 31 to 60 DPD removed 71% of headroom	Chapter 11; T07
Currency	20% a year depreciation turns a 46.4% Base IRR into (16.0%); no price increase on new contracts gives 7.9%	Default sensitivities
Volume	20% lower volume lowers the Base IRR from 46.4% to 37.8%, with no covenant breach	Default sensitivities

Model defaults and scenarios

Product tiers (fictional market, LCY, opening rate 130 per USD)

Tier	Product	Tenor	Monthly hazard	Collection on paying accounts	Repossession rate	Advance rate
1	Pico solar kit, 10 Wp	12	3.5%	88%	0%	50%
2	SHS with TV, 80 Wp	24	2.6%	88%	20%	70%
3	Large SHS with DC fridge, 300 Wp	30	1.9%	90%	40%	70%
4	Inverter with lithium, about 1.2 kWp	36	1.3%	92%	60%	75%
5	Inverter with lithium, about 3 kWp	48	1.0%	93%	70%	75%

Scenario levers

Lever	Base	Downside	Severe
Default hazard multiplier	1.00	1.30	1.75
Collection rate multiplier	1.00	0.97	0.92
Volume multiplier	1.00	0.90	0.75
Hardware cost multiplier	1.00	1.05	1.10
Local currency depreciation per year	5%	12%	25%

Default results (fictional company, USD 4.0m ticket)

Case	Peak equity (USD m)	Y5 EBITDA margin	Investor IRR	Multiple	Breach months
Base	10.0	21.6%	46.4%	6.7x	0
Downside	10.0	7.3%	(22.9%)	0.3x	44
Severe	46.6	(18.4%)	Total loss	0.0x	48

Credit definitions: default at 180 DPD; Stage 2 from 30 DPD and Stage 3 from 90 DPD for the indicative ECL; borrowing base eligibility to 30 DPD; recovery cost 15% of resale proceeds; cure rate 30% (indicative ECL only); ECL discount rate 20%. Other defaults: inflation 6%, price increase on new contracts 5% a year, FX pass through 50%, tax 30% with unlimited loss carry forward, payment burden threshold 10%.

Covenants and facility terms

Default covenants in the model

Covenant	Threshold	Design note
Trailing 3 month collection rate	At least 70%	Set from the company's own history, with a cure period
Receivables at risk	At most 15%	Definition aligned with current PERFORM documents
30+ DPD	At most 25%	
90+ DPD	At most 18%	
Borrowing base headroom	At least zero	Tested monthly
Debt to book equity	At most 3.0x	Book equity depends on ECL methodology
Annual DSCR	At least 1.20x	Ill suited to a growing book; prefer portfolio covenants

Rules for setting them

1. A covenant breached on the day it is signed is a waiver waiting to be requested. Set the level the book can meet on its own history and step it up as performance improves.
2. Equity cures make sense for leverage covenants, not for collection or PAR covenants: new equity does not change how customers pay.
3. Eligibility beyond 30 DPD adds little in a healthy book and puts advances against the receivables most likely to default.
4. Cap unproven products in the borrowing base until twelve months of cohort data sit within an agreed tolerance of plan.
5. Prefer local currency debt for the receivables book; size dollar debt against the depreciation the company can absorb.
6. Ask for a monthly borrowing base certificate by tier and DPD bucket, and a vintage block in every report.

Terms to settle in a facility term sheet

Borrower and structure; amount and tenor; currency and pricing; borrowing base; eligibility criteria; cash management and sweep; portfolio covenants; corporate covenants; cure periods and triggers; reporting; conditions precedent; events of default (Template T07).

Diligence essentials

Data request: the first ten items

1. Account level loan tape since launch (Template T06 structure).
2. Monthly portfolio data by tier for at least 24 months: buckets, due, collected, write offs, recoveries.
3. Cohort observations by tier at fixed ages.
4. Reconciliation of DPD buckets and gross receivables to the ledger at each month end.
5. Credit, write off, restructuring and repossession policies with version dates.
6. Price plan history by tier, including promotions.
7. Agent register, commissions, clawbacks and sales by agent.
8. Unlock logs separated between full payment and other unlocks.
9. Audited and monthly management accounts and budget.
10. Financing agreements, compliance certificates and waivers.

Red flags

Red flag	Why it matters
DPD buckets that do not reconcile to gross receivables or the ledger	Every ratio built on them is unreliable
Collection rate including deposits	Rises with sales, not with repayment
Restructurings that reset DPD, clustered before reporting dates	Re-ageing hides arrears
Unlocks without matching payment	Weak platform control; ownership overstated
Loss allowance flat while PAR30 rises	Profit overstated
Write off policy changed during the history	Ratios not comparable over time
Recovery assumptions above observed resale proceeds	LGD understated; SolaraPay observed about 99%
Sales concentrated in agents or regions with weak cohorts	Origination quality problem
Commission paid fully at activation, no clawback	Incentive to sign marginal customers
Case that clears the hurdle only with RBF, untested tiers or a high exit multiple	Return rests on assumptions outside the company's control
Cohort data not produced within two weeks	The company is unlikely to be managing to it

Decision map

Question	Book	Model	Template	Tool
What business am I in?	1	Dashboard, Unit_Economics	T08 canvas	D2
Can the customer pay?	2	Consumer_Risk	T03 credit policy	D1
What could stop me operating?	3	Inputs, Scenarios	T02 checklist	
Which price plan?	4	Products		D1
What does a customer cost?	5	Products, Unit_Economics	T04 agent economics	D2
How will customers pay?	6	Curves, Vintage_Dashboard	T05 Vintage	D3
Is the portfolio healthy?	7	Credit_Portfolio	T05 lender report	
What profit am I making?	8	FS, Credit_Engine		
Does each sale create value?	9	Unit_Economics		D2
Why do we run out of cash?	10	FS, Financing		D4
How should growth be financed?	11	Financing, Covenants	T07 borrowing base	D4
Is a securitisation feasible?	12	Inputs (structure 2)	T07 term sheet	
How should subsidies be used?	13	RBF_Engine		
What can kill the company?	14	Scenarios, Sensitivity		D4, D6
Would I invest or lend?	15	Valuation, Investment_Readiness	T01 memo, T02 checklist	D5, D6

The model in twelve steps

Step	Sheets	Key action	Check before moving on
1 Start	Cover, Contents, Checks	Save a copy; set currency and first month	Master check OK
2 Business	Products, Consumer_Risk	Catalogue, segments, surveyed incomes	Tier labels match capacity
3 Market	Inputs, Scenarios	FX, inflation, price increase, pass through	Depreciation not set to zero
4 Demand	Inputs, Products	Units by year; mix	Mix totals 100%
5 Revenue	Products, RBF_Engine	Price plans; RBF design	Daily rate not entered as monthly
6 Hardware and capex	Products, Inputs	FOB cost in USD, duty, installation	No double count of FX
7 Operating costs	Products, Inputs	Commission, marketing, servicing, central costs	Central costs sized for the volume
8 Financing	Inputs, Credit_Assumptions	Equity, term loan, facility, eligibility	Borrowing base, not limit, binds
9 Statements	FS, Annual	Read balance, revenue mix, credit losses, cash	Balance check zero every month
10 Scenarios	Scenarios, Sensitivity	Base, Downside, Severe	Selector back to Base before saving
11 Bankability	Credit_Portfolio, Covenants, Credit_Input, Vintage_Input	Load actual history; recalibrate	DPD buckets reconcile
12 Memo	Valuation, Unit_Economics, Benchmark_Compare, Investment_Readiness	Valuation, benchmarks, readiness	Never "investment grade"

Investment memo and readiness

Memo structure (Template T01)

1. Recommendation and conditions.
2. Company, market and products.
3. Unit economics by tier.
4. Portfolio quality and data.
5. Projections and funding requirement.
6. Scenarios and stress tests.
7. Financing structure, borrowing base and covenants.
8. Valuation and returns.
9. Benchmarking and calibration.
10. Consumer protection and impact.
11. Readiness gates and outstanding diligence.
12. Risks and mitigants.

Writing conditions

Type	Use for	Example
Condition precedent	What must be true before money moves	Reconciled data tape; board approved pricing policy
Covenant	What must stay true	Trailing collection rate with cure period
Structural feature	How risk is shared	Tranches tied to cohort performance; valuation ratchet; tier caps
Post closing undertaking	What must be built	Ownership tracking; independent ECL review

Readiness gates

The model's Investment_Readiness sheet holds 23 gates. Automatic gates test integrity, covenant compliance, unit contribution, actual credit data, consumer protection evidence, outcome linked RBF support and data reconciliation. Manual gates record management sign off, term sheets, legal review, the Excel test and auditor review of ECL. The banner never claims investment grade; at most it reads "ready for independent validation". Default inputs meet 3 gates; the SolaraPay case meets 5.

SolaraPay at a glance

Item	Value
Company	SolaraPay Ltd, Republic of Kivara (fictional); currency KVS, 135 per USD at entry
Ask	USD 4.0m equity at USD 8.0m pre money (33.3% stake); KVS 4.5bn receivables facility at 16%
History (synthetic)	Collection rate 77.4% in year 1, 69.3% in year 2; latest month 66.5%; PAR30 17.0%; LGD proxy about 99%
Cohorts at M12	Tier 1 64.6% against 70.3%; Tier 2 69.4% against 74.5%; Tier 3 76.3% against 79.6%
Recalibration	Hazards × 1.30 for Tiers 1 to 3; collection rates down 1 to 2 points
Calibrated Base	Revenue USD 8.1m to 51.2m; EBITDA margin 16.4% in Year 5; EBITDA positive from month 29
Returns	IRR 29.0%, multiple 3.6x (management plan 33.1%, 4.2x); Downside total loss
Mitigants	Full FX price indexation lifts the Downside to 23.6%; entry at USD 6m pre money lifts the Base to 33.8%
Lender view	Lowest projected trailing collection rate 70.0% against a 70% covenant; proposed 65% with cure period
Recommendation	Conditional Go with seven conditions; 5 of 23 readiness gates

Glossary

Term	Meaning
Advance rate	Share of eligible receivables a lender funds
Borrowing base	Maximum facility drawing allowed by eligible receivables and advance rates
CAC	Customer acquisition cost; narrow (commission and marketing) or fully loaded (with field organisation)
Cohort or vintage	Accounts originated in the same month, for one tier
Cure	Return of a delinquent account to current status
Daily rate	Price of one day of service, prepaid by mobile money
Deposit	Upfront payment that activates the device and screens the customer
DPD	Days past due, as schedule shortfall
ECL	Expected credit loss
First loss	Tranche or equity that absorbs losses before any other investor
Hazard	Monthly share of paying accounts that stop paying for good
LGD, PD	Loss given default; probability of default (indicative proxies in the model)
Lockout	Remote disabling of a device when credit runs out
LTV to CAC	Lifetime contribution before CAC ÷ CAC; undiscounted
PAR30	Portfolio more than 30 days past due
RBF	Results based financing
SICR	Significant increase in credit risk (IFRS 9)
True sale	Transfer of receivables to an SPV that removes them from the originator's estate in insolvency
Unlock	Permanent release of the device when the plan is fully paid

Status of external figures

Sector and company figures quoted in Volume 2 rest on secondary reporting and are listed with their status in Annex G of the book. None has yet been checked against its primary document; the M-KOPA FY2024 figures conflict between sources and are provisional. Check the original before quoting any of them. Nothing in this guide is investment, legal, tax or accounting advice.